# Tushin Kulshreshtha

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## Summary


Software engineer and CTO focused on **system architecture**, full-stack products, backend services, **workflow automation**, and building useful harnesses for **AI and agent systems**. Experienced with Python, JavaScript, React, Next.js, PostgreSQL, Docker, AWS, and secure REST APIs.


## Skills

**Languages:** Python, Java, JavaScript, C, C++
**Frontend & Backend:** React, React Native, Next.js, Node.js, Spring Boot, Django, REST APIs
**Databases & Cloud:** PostgreSQL, MySQL, MongoDB, AWS, Azure, Firebase, Docker
**Tools & Systems:** Git, Linux, Temporal, n8n, OAuth 2.0, JWT, OWASP, TensorFlow, PyTorch

## Experience

**US Neuro** **Present**
*Software Engineer*

- Designed, built, and independently maintained the company's **first end-to-end automation system** for position-statement preparation, reducing hands-on preparation time from approximately one hour to **2–3 minutes** while preserving human review.
- Built Azure-based **OCR and document-processing workflows** for checks and EOBs, automating data extraction and supporting final-payment determination while reducing manual reporting work by approximately **90%**.
- Developed an internal Azure-backed **HR workflow application** for equipment and bonus requests, including multi-level approvals and end-to-end request tracking.
- Scaled automated case-processing capacity from approximately **10 to 100 cases per day** through a system I architected and operated independently, supporting a business workflow associated with approximately **$3M in monthly profit**.

**Newtonite**
*Chief Technology Officer & Co-Founder* *Remote*

- Developed Newtonite's **primary architecture** and Cluzar product foundation across AWS and Firebase, defining application structure, service boundaries, data flows, and deployment strategy.
- Designed the platform's engineering foundation to support **maintainable product development**, scalable services, and reliable integration of new capabilities.
- Owned **technical direction from architecture through implementation**, balancing product requirements, engineering tradeoffs, and long-term system evolution.

**SellWizr** **June 2025 – December 2025**
*Software Engineer Intern* *New York, NY*

- Engineered Docker-based **microservices infrastructure** for an enterprise CRM, reducing deployment time by **40%** and improving scalability.
- Designed AI workflow systems with Temporal and n8n, improving workflow optimization by **60%**.
- Implemented test-driven development in agile cycles, reaching **98% code coverage** and reducing production defects by **70%**.

**Site Service Software Inc.** **Aug. 2024 – Apr. 2025**
*Software Engineer Intern* *New York, NY / San Francisco, CA*

- Spearheaded **full-stack migration** to React, Spring Boot, and PostgreSQL, modernizing legacy systems and improving performance by **40%**.
- Implemented OAuth 2.0, JWT management, and OWASP-compliant security controls, reducing vulnerabilities by **85%**.
- Developed **20+ RESTful APIs** with validation and query optimization, reducing response times by **30%**.

**MeetMux** **Feb. 2025 – June 2025**
*AI/ML Engineer* *Remote*

- Architected a hybrid recommendation system with TensorFlow and PyTorch, achieving **95% accuracy** in preference prediction for **50K+ concurrent users**.
- Implemented deep-learning cold-start algorithms, increasing new-user engagement by **40%** and session duration by **25%**.
- Optimized model inference with XGBoost and Scikit-learn, achieving **60% faster inference times**.

**Glitter** **January 2023 – December 2023**
*Lead Software Engineer* *San Francisco, CA*

- Led software engineering efforts across development teams, guiding **technical architecture** and implementation decisions for product systems.
- Coordinated technical planning, code quality, and delivery across the engineering team while translating product needs into **maintainable software**.

## Projects

**Dex Bridge** — [GitHub] — **Python**, **Qdrant**, **MCP**, Docker

- Built a local-first MITM system with mitmproxy to intercept, parse, deduplicate, and capture AI conversations across providers.
- Integrated vector embeddings, Qdrant, Docker, and the Model Context Protocol to provide persistent semantic search over conversation memory.

**File System Project** — [Project] — **C**, **Make**, **GCC**, Docker

- Implemented a Unix-like shell and virtual disk file system in C with create, read, write, delete, directory, and path-resolution operations.
- Built FAT-based free-space management with a Volume Control Block, Linux file transfer, command history, help tooling, and libreadline integration.

**Auto Reel Creation** — [GitHub] — **n8n**, **Remotion**, Docker, Claude

- Built an automated n8n and Remotion pipeline that researches AI developer tools, generates evidence-backed dialogue and visuals, and renders 1080x1920 reels with local voices and word-highlighted subtitles.
- Engineered multimodal production gates for visual relevance and dialogue alignment, local-model fallback, Dockerized services, GitHub Actions deployment, and a live operations dashboard.

## Education

**San Francisco State University** **December 2025**
*Bachelor of Science in Computer Science — GPA: 3.95 — Phi Beta Kappa Honor Society* *San Francisco, California*
**Relevant Coursework:** Data Structures, Algorithms, Operating Systems, Databases, Computer Networks